

Chapter 10 Study Guide

Chapter 10 – Gas Laws and Kinetic Molecular Theory

10.1. Properties of Gases

1. Know the properties of a **gas** that distinguish it from the condensed matter phases of **solids** and **liquids**.
2. Know that **pressure** is defined as a force exerted on an area.
3. Know that a **barometer** is a device for measuring the absolute pressure by determining the height of a column of mercury (mm Hg) rising in an evacuated tube.
4. Know that a manometer is a device for measuring the **relative pressure**.
5. Know that the SI pressure unit is the **pascal (Pa)** which is equal to a newton per square meter (N/m^2) and be able to convert between different pressure units given the conversion factor.

10.2. Historical Gas Laws

1. Be able to use **Boyle's law**, $P_1V_1 = P_2V_2$, to calculate the value of one of the variables given values for the other three variables under conditions of constant temperature and amount of gas.
2. Be able to use **Charles's law**, $V_1/T_1 = V_2/T_2$, to calculate the value of one of the variables given values for the other three variables under conditions of constant pressure and amount of gas recognizing that temperature (T) must be in Kelvin units.
3. Be able to use **Avogadro's law**, $V_1/n_1 = V_2/n_2$, to calculate the value of one on the variables given values for the other three variables under conditions of constant pressure and temperature.

10.3. The Combined and Ideal Gas Laws

1. Be able to use the **combined gas law**, $P_1V_1/T_1 = P_2V_2/T_2$, to calculate the value of one on the variables given values for the other five variables under the condition of constant amount of gas.
2. Understand that an **ideal gas** is characterized by consisting of particles that experience neither attractive nor repulsive interactions with each other, and thus no truly ideal gas exists. However, near room temperature and 1 atm pressure the small attractive and repulsive interactions of real gases can be neglected so most real gases can be well approximated as ideal gas under these conditions.
3. Recognize that ratio of pressure time volume divide by the amount of gas times the temperature (PV/nT) for an ideal gas is constant, referred to as the **ideal gas constant**, and denoted by the symbol R .
4. Be able to use the ideal gas law, $PV = nRT$ to calculate the value of P or V or n or T given the value of three of these quantities, recognizing that the temperature (T) must be in Kelvin units.
5. Recognize that $n = m/M$ (mass divided by formula weight) and replacing n in the ideal gas law with m/M gives after rearrangement

- a. Gas formula weight, $M = mRT/PV$
- b. Gas density, $d = m/V = PM/RT$

10.4. Partial Pressure and Gas Law Stoichiometry

1. Be able to explain and apply **Dalton's law of partial pressure** especially in the common laboratory procedure of collecting a gas over water.
2. Know the definition of **mole fraction** of a component A in a mixture, $\chi_A = n_A/n_{\text{total}}$, and that the partial pressure of a gas, P_A , is related to its mole fraction through $P_A = \chi_A P_{\text{total}}$.
3. Be able to perform stoichiometric calculations with gaseous reactants and/or products where the amount of the gas is specified by P , V , and T values allowing the moles of gas to be determined by $n = PV/RT$.